

# **(TR-104) PYTHON WITH AI/ML**

## **Training Day 12 Report**

**Date: 27 January 2026**

### **Topic Covered: NumPy Library – Array Operations, Vectorization & Data Manipulation**

The twelfth day of training focused on exploring the NumPy library and performing advanced array-based operations using vectorization, slicing, and mathematical transformations. The session included solving multiple practical tasks to strengthen understanding of efficient data processing techniques.

- Created and manipulated arrays using NumPy.
- Developed an 8×8 checkerboard pattern using slicing only.
- Generated a 5×5 random matrix and applied normalization using mean and standard deviation (Z-score normalization).
- Created a 10×10 array and implemented border control by setting internal values to zero.
- Performed in-place conditional filtering by replacing odd numbers with -1.
- Found the nearest value to a target in a 1D array and returned its index.
- Calculated a 3-day moving average using slicing and cumulative sum concepts.
- Worked on image-style array manipulation using a 100×100×3 array, including cropping and horizontal flipping using slicing only.
- Practiced vectorization techniques to improve performance and avoid loops.

#### **GitHub Link for the Tasks:**

[https://github.com/JaspinderKaurWalia26/Python\\_Practice\\_Codes/tree/main/27-01-2026](https://github.com/JaspinderKaurWalia26/Python_Practice_Codes/tree/main/27-01-2026)

Overall, the session helped in strengthening understanding of NumPy arrays, vectorized operations, slicing techniques, and their application in efficient data processing and real-world computational tasks.